Spacecraft Motion Simulation Using Cable-Driven Parallel Robots

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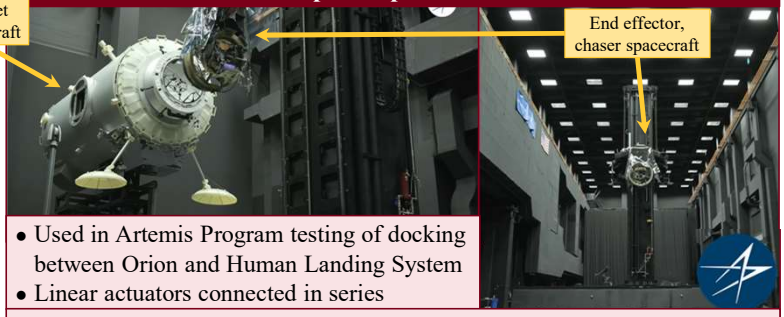
Motivation and State of the Art: Motion Simulation

Real World Applications



- Rendezvous, proximity operations, and docking
- On-orbit servicing, assembly, and manufacturing
- Difficult to simulate and test software and hardware on Earth

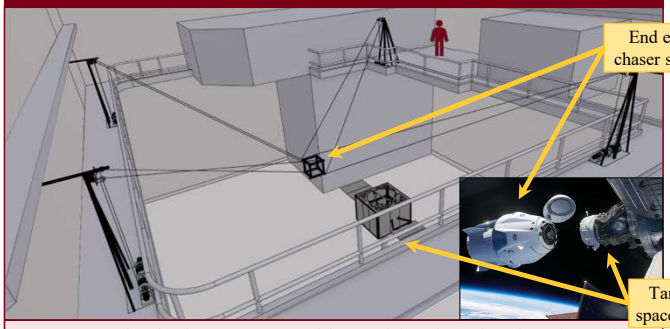
Lockheed Martin's Space Operations Simulation Center



- Used in Artemis Program testing of docking between Orion and Human Landing System
- Linear actuators connected in series
- Pros: rigid, high weight capacity, easier estimation and forward kinematics
- Cons: expensive, heavy, limited DoF and workspace

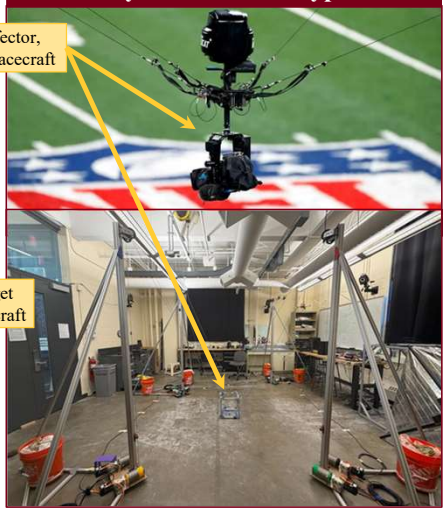
The Alternative: Cable-Driven Parallel Robots (CDPRs)

SPACESiM



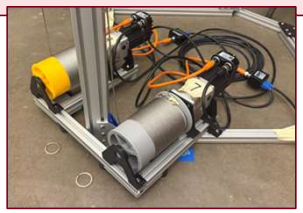
- Spacecraft Platform Actuated by Cables for Simulated Motion
- Hardware- and software-in-the-loop testing
  - Guidance, navigation, and control algorithms
  - Camera, LiDAR, radar, IMU, horizon sensor, etc.
- 40' x 40' x 40' room with 100 lb capacity
- In collaboration with Cornell and NYCST

Skycam and Prototype



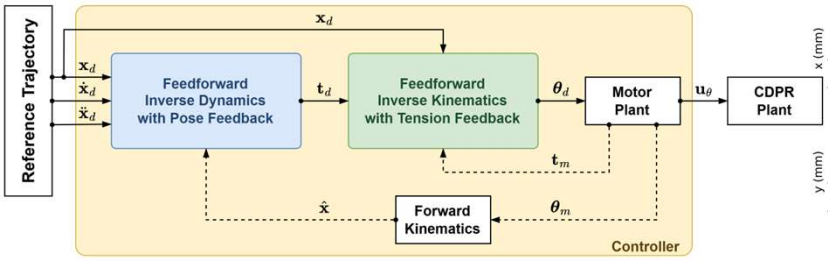
CDPRs

- End effector driven via cables and winches
- Example: Skycam, a four-cable CDPR
- Pros: 6-DoF, lightweight, low inertia, high acceleration, large workspace
- Cons: cable sag, challenging estimation and forward kinematics



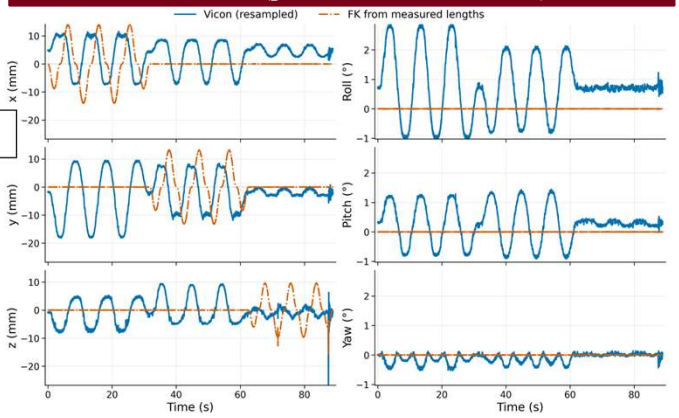
Ongoing Work: Control and Estimation Algorithms

Control Algorithm Sketch



- Goal: reduce pose tracking error with nested tension and pose feedback
- Challenges: noisy tension (motor torque) readings; real-time deterministic execution at 3 ms or less

Pose Error (planned minus observed)



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